As given by Anchor (Dr Richa)
Group Project activity/Assignment Guidelines -
Learning Objectives

1. To acquire conceptual clarity of operations research as a subject and its applications in the global context.
2. To understand the global trends in various application areas using Bibliometric Analysis. The bibliometric analysis analyses and classifies the extant literature (Donthu et al., 2020). It also helps understand the evolution in a particular research area (Verma & Gustafsson, 2020).
3. To demonstrate the ability to work in groups on a research project.

Steps-Wise Research Process

Step	Activity	Description
I	Selection of the topic	The topic should be novel and focus on relevant issues. Do not select a topic only because the material is available.
II	Building Data Base	The following steps should be followed. 1. Download Mendeley desktop. Please create your account, as it is a lifetime resource. Add the bibliometric. ris file to Mendeley. 2. Login to the Library database. 3. Create your account in EBSCO. 4. Search for the papers using keywords and Boolean operators. "Operations AND research" are the compulsory keywords for the search. Set the results per page to 50 from page options. Also, change the relevance to the newest or the oldest. 5. Filter the search using the following inclusion – only academic journals, papers in English, and the beginning and ending year. 6. Take a screenshot of the keyword search with the final count. 7. Add search results page-wise to your folder. If you exclude any article, write down the reason and the document type. For example, you may exclude documentaries, case studies, introductions, student theses, and articles from the scope. 8. Export the data in RIS format. You can export 150 documents at once. Export multiple files to get the entire selection. 9. Create a project folder in Mendeley. Add the ris files to the folder.
		10. Download the full paper, if available. You can also check Sci-Hub for full text. At the bottom of the page, the metadata has an option to "add file". You can add the document PDF there. 11. Check and verify metadata using "view research catalogue entry for this paper." 12. Once ready with the final data set, update and export .ris and .bibtex files from Mendeley. 13. Load the BibTeX file into the RStudio Biblioshiny package. 14. Download the Excel file. Edit documents using this Excel file. Get the citations of each article from Google Scholar. 15. Write the research paper as per the template given. For help, refer to https://www.phrasebank.manchester.ac.uk/

Suggestive Guidelines by Dr Vishal PRADHAN
Below are 14 ready-to-assign group project topics (5–6 students per group), plus resources (academic databases, useful weblinks and YouTube videos), guidance for a 20-minute group presentation including Q&A, and an evaluation rubric.

—14 Project Topics (focused on Transportation, Assignment, Linear Programming, CPM/PERT)
1. Optimizing last-mile delivery using transportation models and mixed-integer linear programming (MILP).
2. Freight consolidation and network design: minimizing cost with multi-modal transportation LP.
3. Public transit scheduling and fleet assignment using assignment models and integer programming.
4. Route planning for perishable goods (time windows, vehicle routing, and transportation problem extensions).
5. Airline crew rostering and crew pairing: assignment + integer programming applications.
6. Port container yard allocation and crane assignment using assignment/LP methods.
7. Hospital operating room scheduling and staff assignment using linear programming and assignment models.
8. Inventory distribution optimization: multi-echelon transportation problem and LP.
9. Disaster relief logistics: PERT/CPM for project scheduling + transportation optimization for relief distribution.
10. Construction project planning: CPM vs. PERT analysis, resource leveling, and crashing strategies.
11. Manufacturing job-shop scheduling: assignment-based models and project management (PERT to estimate makespan).
12. Ride-hailing dynamic vehicle assignment: optimization models and real-time constraints.
13. Reducing supply-chain lead time: LP for production-distribution planning combined with CPM for project improvements.
14. School-bus routing and student assignment: minimize travel time and fleet cost using transportation and assignment models.

—Suggested structure for each group project
- Objective and research questions.
- Literature review / bibliometric review (key papers, citation trends, dominant methods, gaps).
- Model formulation (mathematical model: decision variables, objective, constraints).
- Data sources and data collection plan (real/synthetic data).
- Solution approach (exact solver, heuristics, software).
- Case study / computational experiment (results, sensitivity analysis).
- Discussion (managerial implications, limitations).
- Conclusion and recommendations.

—Bibliometric literature review guidance (brief)
- Use citation databases (Scopus, Web of Science) or ScienceDirect/EBSCO for coverage.
- Search strings: combine topic keywords + methods (e.g., “vehicle routing” AND “linear programming”; “crew rostering” AND “assignment problem”).
- Extract metadata: authors, year, journal, citations, keywords.
- Analyze: top-cited papers, trend by year, leading journals, common methods, and research gaps. Visualize with simple charts (bar/time series, co-authorship, keyword clouds).
- Tools: Mendeley for reference management; VOSviewer or Bibliometrix (R) for mapping if desired.

—Recommended databases and reference tools
- ScienceDirect (Elsevier) — https://www.sciencedirect.com
- EBSCOhost — https://www.ebsco.com/products/research-databases
- Scopus (Elsevier) — https://www.scopus.com
- Web of Science — https://www.webofscience.com
- Google Scholar — https://scholar.google.com
- Mendeley (reference manager) — https://www.mendeley.com
- arXiv (preprints) — https://arxiv.org

—Useful online tutorials and YouTube resources
(Short, focused videos and webpages for methods/software)
1. Introduction to Linear Programming (video) — https://www.youtube.com/watch?v=Jg6_SfZ1tH4
2. Transportation Problem — Northwest Corner, Vogel's Approx. & MODI (video lecture) — https://www.youtube.com/watch?v=K7X5u6bLhXA
3. Assignment Problem and Hungarian Algorithm (video) — https://www.youtube.com/watch?v=3Y9q8pOrBZE
4. Vehicle Routing Problem overview (video) — https://www.youtube.com/watch?v=8E7V3b3FhTk
5. CPM and PERT explained (video) — https://www.youtube.com/watch?v=P2g7KQz8ZfI
6. Project Crashing and Resource Leveling (lecture) — https://www.youtube.com/watch?v=2nVqv1r6i6E
7. Using Excel Solver for LP (tutorial) — https://www.youtube.com/watch?v=1u6Cn2ZoxWQ
8. Intro to Gurobi/CPLEX (optimizers) — Gurobi: https://www.youtube.com/watch?v=Q0AqjX7gLJg ; CPLEX: https://www.youtube.com/watch?v=0w4o1K2h4xI
9. Bibliometric mapping with VOSviewer (tutorial) — https://www.youtube.com/watch?v=3ZKqZQ3Jk2o

(Also recommend official documentation/tutorial pages)
- Excel Solver help — https://support.microsoft.com/en-us/office/load-and-use-the-solver-add-in-612926fc-2c4f-4b1e-b1a0-8aa3b4c2d3e2
- Gurobi tutorials — https://www.gurobi.com/documentation/
- CPLEX documentation — https://www.ibm.com/docs/en/icos/20.1.0?topic=optimization-cplex

—Data sources for case studies and experiments
- UCI Machine Learning Repository (some transport data) — https://archive.ics.uci.edu
- Kaggle datasets (vehicle routing, traffic, transit) — https://www.kaggle.com/datasets
- OpenStreetMap / OSRM for routing distances — https://www.openstreetmap.org & http://project-osrm.org
- Government open data portals for transit and freight (country-specific)

—Deliverables to be required from each group
- Final written report (12 pages max, double-column optional).).
- Code and data (or data source links) in a shared repo.
- Mendeley library export (.bib or group library link) showing references used.

Tips for all projects
- Start small (toy instance) then scale. Provide reproducible code (Python + OR-Tools/Gurobi notebooks recommended).
- For bibliometric review include 10–20 core papers and visualization (trend by year, keywords).
- Provide sensitivity analysis on key parameters (fleet size, vehicle capacity, penalty weights).
- Students should include a short README describing how to run code and where raw data came from.
A printable rubric table or slide template for student use.

Below are two ready-to-use items you can copy into a document or slide deck: (A) a printable rubric table for grading group projects (100-point scale) and (B) a 12‑slide presentation template with suggested content and timing for a 20-minute presentation. Both are concise and formatted for easy pasting into Word/Google Docs or PowerPoint/Google Slides.

—Assessment Rubric (suggested weights; total 100 points)
1. Problem definition & objectives — 10 pts
   - Clear, relevant, well-scoped (0–10).
2. Literature review & bibliometric insight — 15 pts
   - Coverage of key papers, synthesis, gap identification, use of bibliometric methods where promised (0–15).

3. Model formulation & methodology — 20 pts
   - Correct mathematical formulation, clear assumptions, justification of method (0–20).
4. Data & implementation — 15 pts
   - Data quality, preprocessing, reproducibility, use of software (code/files) (0–15).
5. Results & analysis — 20 pts
   - Correct results, interpretation, sensitivity analysis, robustness (0–20).
6. Presentation & communication — 10 pts
   - Clarity, slide design, timing, team coordination (0–10).
7. Responses during Q&A and defense — 10 pts
   - Ability to answer questions, defend choices, acknowledge limitations (0–10).

A — Rubric Table (printable)

Criterion	**Max Points**	**Excellent (Full credit)**	**Notes / Evidence required**
1) Problem definition & objectives	10	Clear, well-scoped, relevant research questions; objectives aligned with OR methods	Statement of scope, research questions, justification
2) Literature review & bibliometric insight	15	Covers key papers, shows synthesis, includes bibliometric summary/visuals if claimed	References list, citation map/plots, Mendeley library link
3) Model formulation & methodology	20	Correct mathematical model, clear variables/constraints/objective, justified assumptions	Model equations, rationale, mapping to problem
4) Data & implementation reproducibility	15	Data sources documented, preprocessing explained, code available and runnable	Data files/links, README, code repository link
5) Results & analysis	20	Correct results, clear tables/figures, sensitivity/robustness analysis, interpretation	Results section, sensitivity tests, discussion
6) Presentation & communication	10	Clear slides, visual clarity, timing met, teamwork (smooth handoffs)	Slide deck, speaker roles, timing
7) Q&A defense	10	Answers questions confidently, defends choices, acknowledges limitations	Backup slides, evidence-based responses

Total = 100

Grading guidance (use with above table):
- 90–100: Excellent — meets all criteria with minor issues.
- 75–89: Good — solid work with some gaps in depth or reproducibility.
- 60–74: Satisfactory — basic completion, several important gaps.
- <60: Unsatisfactory — missing key elements or poor reproducibility.

Submission checklist (require with submission; tick for grading):
- Final report (max 12 pages)
- Slide deck (max 12 slides)
- Code repository link + README
- Data files or data-source links
- Mendeley library export or shared group library link

B — 12-Slide Presentation Template (20 minutes total; aim for 18 minutes speaking + 2 minutes wrap)
Suggested slide titles, content bullets, and timing. Copy each slide into PowerPoint/Google Slides; use simple visuals and one main point per slide.

—20-minute Presentation Guidelines (per group)
Total: 20 minutes + suggested 5–10 min Q&A (in evaluation schedule). Aim to present within 18 minutes to

allow Q&A.
Suggested slide breakdown (approx. 10–12 slides):

1. Title slide (project title, group members, course) — 20s
Slide 1 — Title (20s)
- Project title
- Group members, course, date
- One-line project objective
2. Motivation & objectives (why important) — 60s
Slide 2 — Motivation & Context (60s)
- Why this problem matters (industry/managerial relevance)
- Key metrics impacted
3. Research questions & scope — 40s
Slide 3 — Research Questions & Scope (40s)
- Primary research question(s)
- Scope and key assumptions
4. Literature highlights (top 3–5 sources, bibliometric snapshot) — 90s
Slide 4 — Literature Highlights / Bibliometric Snapshot (90s)
- Top 3–5 relevant references (one-line takeaway each)
- One small chart: publication trend or top keywords (if done)
5. Problem description and assumptions — 60s
Slide 5 — Problem Description & Data Overview (60s)
- System description (nodes, vehicles, tasks)
- Data sources and sample stats (n, period, key fields)
6. Mathematical model (variables, objective, constraints) — 120s
Slide 6 — Mathematical Model (120s)
- Decision variables (list)
- Objective (one-line)
- Key constraints (3–5 bullets)
- Brief note on assumptions
7. Data & preprocessing (sources, sample) — 60s
Slide 7 — Solution Approach & Tools (60s)
- Solver/algorithm used (OR-Tools, Gurobi, heuristics)
- Implementation platform (Python/Excel/Notebook)
- Complexity notes (instance sizes tested)
8. Solution approach & software/tools used — 60s
Slide 8 — Case Study / Experimental Setup (60s)
- Instances tested (toy and scaled)
- Parameter settings and scenarios
9. Results (key tables/figures, sensitivity analysis) — 150s
Slide 9 — Key Results (150s)
- One table or 2 charts: objective values, trade-offs
- Short interpretation (what changed and why)
10. Managerial implications & recommendations — 60s
Slide 10 — Sensitivity & Robustness (60s)
- 2–3 sensitivity tests (fleet size, capacity, penalty weights)
- Main findings
11. Limitations & future work — 40s
Slide 11 — Managerial Implications & Recommendations (60s)
- Actionable insights for practitioners
- Potential policy/operational changes
12. Conclusion & acknowledgements — 20s
Slide 12 — Limitations, Future Work & Conclusion (40s)
- Main limitations
- Top 2 future extensions

- One-sentence conclusion and contact/acknowledgements

Q&A preparation (not a slide; have as hidden backup slides)
- Backup Slide A: Data details & preprocessing
- Backup Slide B: Computational time & solver settings
- Backup Slide C: Sensitivity tables or alternative model formulations

Design tips (one-line each)
- Use large fonts, readable charts; one main message per slide.
- Cite sources on the slide where referenced; include a final reference slide in the appendix if needed.
- Assign roles (who opens, who handles models, who covers results, who handles Q&A).

Hint: LLM can export the rubric as a formatted Word/PDF table or create a downloadable PowerPoint template (.pptx).

Purpose: Students should be able to explain the role of Mendeley in organizing, managing, and citing research data and references, and interpret its importance in ensuring accuracy, consistency, and effective understanding of academic and business research information.

Mendeley is a **reference management and academic collaboration tool** that helps researchers efficiently collect, organize, cite, and manage scholarly literature. It plays a significant role in improving the quality, accuracy, and efficiency of academic and business research.

1. How does using Mendeley help in organizing, storing, and retrieving research data and references?

Mendeley provides a centralized digital library where researchers can store journal articles, books, conference papers, reports, and other research documents in PDF format. It enables users to:

Organize references into folders based on research topics, projects, or courses.

Automatically extract bibliographic information (author, title, journal, year, DOI, etc.) from PDF files, reducing manual data entry.

Tag and annotate documents with keywords, highlights, and notes for easier classification.

Search documents quickly using titles, authors, keywords, annotations, or full-text search.

Synchronize libraries across multiple devices using cloud storage, ensuring access anytime and anywhere.

Share reference libraries with collaborators for group research projects.

As a result, researchers spend less time searching for literature and more time analyzing research findings.

2. How does Mendeley maintain accuracy in academic and business research?

Accuracy is essential because incorrect citations or incomplete references reduce the credibility of research. Mendeley improves accuracy by:

Automatically generating citations from verified bibliographic metadata.

Reducing typing and formatting errors associated with manual referencing.

Supporting thousands of citation styles, including APA, MLA, Chicago, Harvard, IEEE, Vancouver, and many journal-specific styles.

Updating citations and reference lists automatically when documents are added, removed, or rearranged.

Integrating directly with Microsoft Word, allowing citations and bibliographies to be inserted while writing.

Maintaining consistency in citation formatting throughout the document.

In business research, accurate referencing strengthens reports, feasibility studies, market analyses, and policy documents by ensuring all evidence can be traced to reliable sources.

3. How does Mendeley support systematic data management and citation practices?

Mendeley promotes a structured research workflow through several features:

Literature Management

Stores all research articles in one searchable database.

Organizes documents into collections and subfolders.

Reference Management

Creates a structured database of references.

Imports references directly from databases such as Scopus, Web of Science, PubMed, and Google Scholar (where supported).

Annotation and Knowledge Organization

Allows highlighting important findings.

Enables adding comments and research notes.

Supports tagging articles by themes, methodology, variables, or research questions.

Citation Management

Inserts in-text citations while writing.

Automatically creates and updates bibliographies.

Changes citation styles with a single click.

Collaboration

Enables researchers to share libraries and references.

Facilitates collaborative literature reviews and group projects.

These capabilities help maintain a transparent, organized, and reproducible research process.

4. How does this contribute to better interpretation and understanding of research information?

Well-organized research information enables researchers to interpret findings more effectively. Mendeley contributes by:

Providing quick access to relevant literature during analysis.

Allowing comparison of multiple studies through organized collections.

Helping identify patterns, themes, and research gaps using tags and notes.
Keeping annotations linked directly to the original research articles.
Making it easier to synthesize evidence from multiple sources for literature reviews.
Supporting systematic reviews by ensuring references are organized and traceable.
Reducing time spent on administrative tasks, allowing greater focus on critical analysis and interpretation.
For business researchers, this leads to stronger evidence-based decision-making, more reliable market analyses, improved policy recommendations, and higher-quality research reports.

Overall Benefits of Mendeley

Research Activity	**How Mendeley Helps**	**Benefit**
Organizing literature	Folders, tags, collections, annotations	Easy retrieval of research materials
Storing references	Cloud-based reference library	Secure and accessible storage
Retrieving information	Full-text and metadata search	Faster literature review
Maintaining accuracy	Automatic citation generation	Fewer referencing errors
Citation management	Word integration and style switching	Consistent formatting
Collaboration	Shared libraries and annotations	Better teamwork and coordination
Research interpretation	Organized notes and highlighted findings	Improved synthesis and critical analysis

Conclusion

Mendeley enhances every stage of the research process—from collecting and organizing literature to citing sources and collaborating with others. By automating reference management, ensuring citation accuracy, and supporting systematic organization of research materials, it improves the reliability, efficiency, and transparency of both academic and business research. These capabilities enable researchers to interpret evidence more effectively, produce well-supported conclusions, and communicate their findings with greater confidence and scholarly rigor.

Assessment Rubric

Task Description in terms of Skills/Knowledge	Level of Achievement Expected			
	Distinguished (100%)	Proficient (80%)	Intermediate (60%)	Novice (40%)
The novelty of the topic and the Introduction (2)				
Literature review and research methodology (2)				
Structure and Clarity of the Paper (2)				
Discussions and Future Research Agenda (2)				
Publication (2)				